SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 2

Industry Problem Solving Task

COURSE NAME: Deep Learning **COURSE CODE: MLA0408**

COURSE OUTCOME COVERED

CO 1: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. (BL-6)

Assessment Tool Weightage: 40%

QUESTIONS: **Total Marks:25**

Q1. Transfer Learning
A healthcare startup aims to develop an AI-based system for detecting skin cancer from dermoscopic images. Since only a limited number of labeled images are available, design a transfer learning strategy using a pre-trained CNN model. Justify your choice of model, identify which layers should be frozen or fine-tuned, and explain how your design improves classification performance.

Q2. DenseNet and PixelNet
A smart city project requires an automated road-scene analysis system that accurately identifies roads, pedestrians, vehicles, and traffic signs at the pixel level. Design a deep learning architecture using DenseNet and PixelNet to solve this problem. Explain how the proposed architecture improves feature reuse, segmentation accuracy, and computational efficiency.

Q3. Bidirectional RNN and Encoder–Decoder
A multinational customer service company wants to build a multilingual chatbot capable of translating customer queries while preserving contextual meaning. Design an Encoder–Decoder sequence-to-sequence model using Bidirectional RNNs. Illustrate the architecture and justify how bidirectional processing improves translation accuracy.

Q4. BPTT and LSTM
A financial analytics company is developing a model to forecast stock prices using historical market data. Create an LSTM-based prediction system and explain how Backpropagation Through Time (BPTT) is used to train the network. Justify why LSTM is more suitable than a traditional RNN for this application.

Q5. Integrated Deep Learning Solution
A media streaming company plans to automatically generate captions for uploaded videos to improve accessibility. Design an integrated deep learning solution that combines Transfer Learning for visual feature extraction with an LSTM-based Encoder–Decoder architecture for

caption generation. Explain the role of each component and justify how the complete system produces accurate captions.

Code of Conduct:

I M MEGHANADH(192325026) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M MEGHANADH

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understandin g; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

Deep Learning – Exam Answers

Q1. Transfer Learning for Skin Cancer Detection

Proposed Model

Use a pre-trained CNN such as EfficientNet-B0 or ResNet50, trained on a large dataset such as ImageNet.

Architecture

Input Dermoscopic Image → Preprocessing and Data Augmentation → Pre-trained EfficientNet/ResNet → Global Average Pooling → Dense Layer + ReLU → Dropout → Output Layer (Softmax/Sigmoid) → Benign/Malignant Classification.

Transfer Learning Strategy

1. Load the pre-trained model without its original classification layer.
2. Freeze the initial layers because they learn general features such as edges, colors, textures and shapes.
3. Fine-tune the last few convolutional layers because they learn task-specific features.
4. Add new classification layers for skin cancer detection.
5. Use data augmentation such as rotation, flipping, zooming and brightness variation.

Why EfficientNet?

EfficientNet provides high accuracy with fewer parameters and lower computational cost. It is useful when only a limited medical image dataset is available.

Advantages

Reduces overfitting, requires less training data, reuses knowledge learned from millions of images, and fine-tuning helps learn specific skin lesion characteristics.

Conclusion

By freezing early layers and fine-tuning higher layers, the system reuses general visual knowledge while adapting to skin cancer images, improving classification performance.

Q2. DenseNet and PixelNet for Road-Scene Segmentation

Objective

Classify every pixel into categories such as road, pedestrian, vehicle, traffic sign and background.

Proposed Architecture

Input Road Image → DenseNet Backbone → Dense Feature Blocks → Feature Maps from Multiple Layers → PixelNet Multi-Scale Feature Extraction → Pixel-wise Classification Network → Upsampling/Decoder → Semantic Segmentation Output.

Role of DenseNet

DenseNet connects each layer to subsequent layers, enabling feature reuse. A dense connection can be represented as: $x_l = H_l([x_0, x_1, ..., x_{(l-1)}])$.

Role of PixelNet

PixelNet performs pixel-level prediction using features extracted from multiple layers. It combines low-level features for edges and boundaries, mid-level features for object parts, and high-level features for semantic meaning.

Advantages

DenseNet improves feature reuse and gradient flow, while PixelNet improves pixel-level classification using multi-scale information. Together they improve boundary detection, segmentation accuracy and computational efficiency.

Conclusion

DenseNet extracts rich reusable features, while PixelNet uses multi-scale information for accurate pixel-wise classification.

Q3. Bidirectional RNN and Encoder–Decoder for Translation

Objective

Build a multilingual chatbot that translates customer queries while preserving context and meaning.

Architecture

Input Sentence → Embedding Layer → Bidirectional RNN Encoder → Combined Context Representation → Decoder RNN → Attention/Context Vector → Translated Output.

Bidirectional Processing

The forward RNN reads left to right and the backward RNN reads right to left. Their hidden states are combined: h_t = [→h_t ; ←h_t].

Encoder–Decoder Process

The encoder converts the input sentence into contextual representations. The decoder generates the translated sentence one word at a time. Example: “I need help” → “Necesito ayuda”.

Why Bidirectional RNN?

Unlike a normal RNN, a Bidirectional RNN considers both previous and future context. This improves understanding of ambiguous words and sentence meaning.

Conclusion

The Bidirectional Encoder captures context from both directions, while the Decoder generates the translated sequence, improving semantic understanding and translation quality.

Q4. BPTT and LSTM for Stock Price Prediction

Objective

Predict future stock prices using historical market data such as opening price, closing price, high price, low price and trading volume.

Architecture

Historical Stock Data → Data Preprocessing and Normalization → Sequence Creation → LSTM Layer 1 → LSTM Layer 2 → Dense Layer → Predicted Stock Price.

LSTM Components

An LSTM uses a forget gate, input gate, cell state and output gate. The forget gate decides what information to remove, while the input gate decides what new information to store.

Role of BPTT

Backpropagation Through Time trains the LSTM using sequential data: (1) pass the input sequence through the LSTM, (2) predict the stock price, (3) calculate prediction error, (4) propagate the error backward through time steps, and (5) update weights using gradient descent.

Why LSTM?

Traditional RNNs suffer from vanishing gradients and have difficulty remembering long-term information. LSTMs use memory cells and gates to preserve important information over longer sequences.

Conclusion

LSTM is suitable for stock prediction because market data contains temporal dependencies. BPTT trains the network by propagating errors through time.

Q5. Integrated Deep Learning Solution for Video Caption Generation

Objective

Automatically generate meaningful captions for uploaded videos to improve accessibility.

Architecture

Video → Frame Extraction → Pre-trained CNN (ResNet/EfficientNet/InceptionV3) → Visual Features → LSTM Encoder → Context Representation → LSTM Decoder → Generated Caption.

Transfer Learning Component

The pre-trained CNN extracts important visual features such as people, cars, animals, roads and other objects. Transfer learning reduces training requirements and computational cost.

LSTM Encoder

The encoder receives a sequence of frame features and learns temporal relationships and action sequences. For example: person standing → person entering a car → car moving.

LSTM Decoder

The decoder generates a caption word by word, for example: → A → person → is → driving → a → car → .

Why the Complete System Works

The CNN provides visual understanding, the LSTM Encoder captures temporal information across frames, and the LSTM Decoder converts the learned representation into natural-language captions.

Conclusion

Combining transfer learning with an LSTM Encoder–Decoder produces an efficient system that understands video content and generates accurate, meaningful captions.